

Accounting & Performance Ratio Model — Technical Specification

Caterpillar Inc. (NYSE: CAT) | FY2024

Created by: Henry Van Brussel

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Course: BUS-314, Shidler College of Business

1. Problem Statement

Caterpillar Inc. is the world's leading manufacturer of construction and mining equipment, diesel engines, and diesel-electric locomotives, operating across 190+ countries. This model analyzes CAT's FY2024 financial performance (fiscal year ended December 31, 2024) using audited data from the company's 10-K filing with the SEC. FY2023 balance sheet figures serve as prior-year comparators for denominator and average-balance calculations. The objective is to compute and interpret 30+ ratios across five categories — performance, profitability, efficiency, leverage, and liquidity — to assess value creation, asset efficiency, and financial risk. Results feed directly into the Stage 4 AI-assisted interpretation memo.

2. Inputs (Known Variables)

Source / Item	Named Range	FY2024 (\$M)	FY2023 (\$M)
BALANCE SHEET			
Cash & cash equivalents	BAL_cash_marketable_securities_2024	6,889	—
Receivables — trade & other	BAL_receivables_2024 / _2023	9,282	9,310
Inventories	BAL_inventories_2024 / _2023	16,827	16,565
Total current assets	BAL_assets_current_2024	45,682	—
Total assets	BAL_assets_total_2024 / _2023	87,764	87,476
Total current liabilities	BAL_liabilities_current_2024	32,272	—
Total liabilities	BAL_liabilities_total_2024	68,270	—
Long-term debt (after 1 yr)	BAL_debt_long_term_2024 / _2023	27,351	24,472
Total shareholders' equity	BAL_equity_shareholders_2024 / _2023	19,494	19,503
INCOME STATEMENT			
Net sales	INC_sales	64,809	—
Cost of goods sold	INC_cost_goods_sold	43,244	—
EBIT	INC_ebit	14,187	—
Interest expense	INC_interest_expense	1,713	—
Depreciation	INC_depreciation	1,413	—
Net income			
CASH FLOW STATEMENT			
Cash from operations	CASH_operating	14,184	—
Capital expenditures	CASH_capex	(1,702)	—
Cash from financing			
MARKET / ANALYST INPUTS			
Share price (fiscal year-end)	share_price	\$349.58	—
Shares outstanding	shares_outstanding	492M	—
Cost of capital (WACC)	cost_capital	7.90%	—
Tax rate (assumed)	tax_rate	21.40%	—

3. Named Range Conventions

Raw inputs: SOURCE_description_year (e.g., BAL_cash_marketable_securities_2024, INC_sales, CASH_operating). Du Pont cross-reference outputs: RATIO_asset_turnover, RATIO_operating_profit_margin, RATIO_debt_burden, RATIO_leverage. Derived intermediates: startYear_, currentYear_, and avg_ prefixes. All named ranges are defined in the Excel Name Manager; no bare cell references appear in the Ratios tab.

4. Assumptions & Constraints

- Tax rate assumed at statutory 21.4%; not derived from INC_taxes / INC_taxable_income.

- Interest expense on simple annual basis; no sub-period compounding.
- No off-balance-sheet items included; operating leases and pension obligations not capitalized.
- Total capitalization = long-term debt (after 1 yr) + shareholders' equity — excludes short-term debt (\$4.4B) and current LTD (\$6.7B).
- Quick ratio uses trade receivables only; Cat Financial finance receivables (\$9.6B) excluded.
- WACC of 7.9% is an analyst estimate with no formal CAPM build-up.
- Share price (\$349.58) reflects fiscal year-end close; market cap = share_price × shares_outstanding, no diluted adjustment.
- Average balances = simple average of FY2023 and FY2024 period-end figures.

Derived Inputs

Performance | Profitability | Efficiency | Leverage | Liquidity | Du Pont

[illegible]

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■ ■ Efficiency
RATIO_asset_turnover      = INC_sales / startYear_total_assets
Receivables turnover      = INC_sales / startYear_receivables
Avg collection period (days) = startYear_receivables / currentYear_daily_sales
Inventory turnover         = INC_cost_goods_sold / startYear_inventory
Days in inventory          = startYear_inventory / currentYear_daily_cogs
Profit margin              = INC_net / INC_sales
RATIO_operating_profit_margin = currentYear_after_tax_op_inc / INC_sales

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■ ■ ■ Leverage	
LT debt ratio	= BAL_debt_long_term_2024 / currentYear_total_cap
LT debt-equity ratio	= BAL_debt_long_term_2024 / BAL_equity_shareholders_2024
Total debt ratio	= BAL_liabilities_total_2024 / BAL_assets_total_2024
Times interest earned	= INC_ebit / INC_interest_expense
Cash coverage ratio	= (INC_ebit + INC_depreciation) / INC_interest_expense
RATIO_debt_burden	= INC_net / currentYear_after_tax_op_inc
RATIO_leverage	= BAL_assets_total_2024 / BAL_equity_shareholders_2024

Liquidity

NWC to assets = currentYear_nwc / BAL_assets_total_2024

Current ratio = BAL_assets_current_2024 / BAL_liabilities_current_2024

Quick ratio = (BAL_cash Marketable securities_2024 + BAL_receivables_2024) / BAL_liabilities_current_2024

Cash ratio = BAL_cash Marketable securities_2024 / BAL_liabilities_current_2024

ROA (Du Pont) = $\text{RATIO_asset_turnover} \times \text{RATIO_operating_profit_margin}$
 ROE (Du Pont) = $\text{RATIO_leverage} \times \text{RATIO_asset_turnover} \times \text{RATIO_operating_profit_margin} \times \text{RATIO_debt_burden}$

- **Scalar inputs table:** All derived intermediates and averages labeled with named ranges and values.
- **Ratio output table:** 30 ratios in six categories with named-range formula references and numeric results.
- **Du Pont tree:** Decomposition confirming ROA and ROE via `RATIO_*` cross-references to source sections.
- **Balance check:** Total Assets = Total Liabilities + Equity verified for FY2024 (\$87,764M) and FY2023 (\$87,476M).
- **Notes tab:** Assumptions log, data source citation, color-coding legend, and full record of Stage 2 formula corrections.

What was built correctly

The named-range architecture is the model's strongest feature — every input is descriptively named, the Du Pont section cross-references `RATIO_*` ranges rather than bare cell addresses, and the Notes tab logs every formula correction with a plain-language explanation. Balance sheet checks pass for both years. Start-of-year vs. average-based profitability variants are consistently maintained.

What to improve

- Quick ratio excludes Cat Financial finance receivables (\$9.6B) without an inline cell note — embed documentation directly.
- Total capitalization omits short-term debt (\$4.4B) and current LTD (\$6.7B); label the metric explicitly or include all debt.
- Tax rate should be derived as `INC_taxes / INC_taxable_income` rather than manually assumed at 21.4%.
- WACC has no build-up; add a sensitivity table (6%–10%) to stress-test EVA for a CFO audience.
- Layout: freeze panes at category headers, add conditional formatting for out-of-norm ratios, group rows by category.
- Add free cash flow (`CASH_operating – CASH_capex = $12,482M`) to contextualize the \$7.8B share repurchase.

8. Limitations & Next Steps

Excluded: Multi-year trend analysis, peer benchmarking (Deere & Co., CNH Industrial), Cat Financial segment-level adjustments, off-balance-sheet capitalization (operating leases, pension), and scenario/sensitivity analysis.

Stage 4 setup: Section 5's Calculation Flow becomes the AI prompt instruction block. Section 7's Model Review directs the AI to flag the four assumptions most requiring commentary: WACC sensitivity, capitalization scope, quick ratio composition, and tax rate derivation. Section 6's Outputs define the ratio table the AI will interpret to produce a CFO- and investor-facing narrative on CAT's value creation, operational efficiency, capital structure risk, and liquidity position.

Data source: Caterpillar Inc., 10-K FY2024, SEC EDGAR. All figures in millions of USD unless noted. Model built in Microsoft Excel with named-range conventions.